StarkCore Sub6 outdoor thermal simulation (12L)

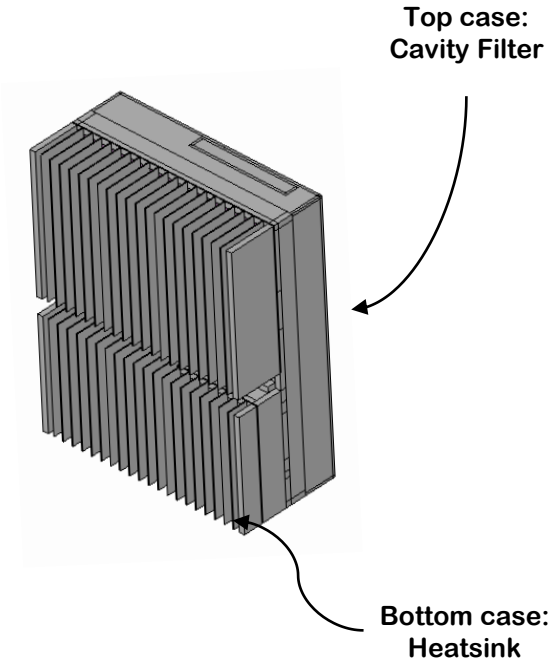
Note:
Customer :

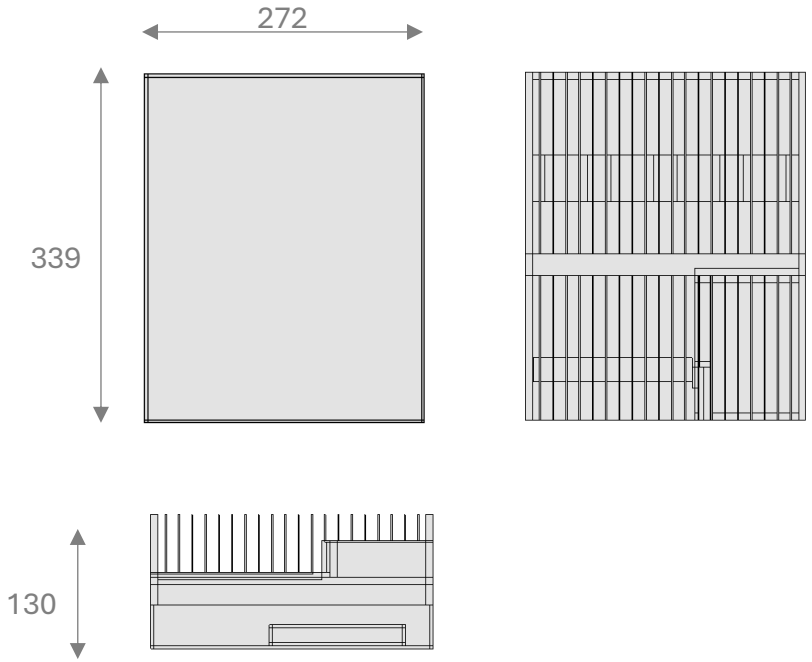
Tedus.chen | AMSBD | 02/12/2026

Boundary Conditions / Dimension / Structure

- Flow type: Natural Convection (Fanless)
- Ambient Temperature: 45 ˚C
- Installation : Pole mount
- Product size (Unit: mm) :

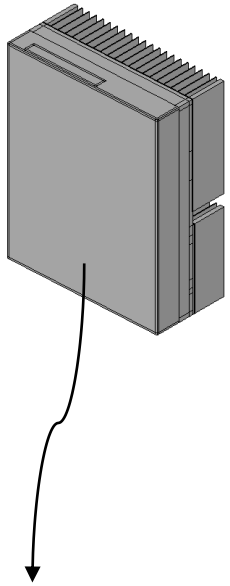
  339(L)*272(W)*130(H) = 11.98L (include waterproof structure)



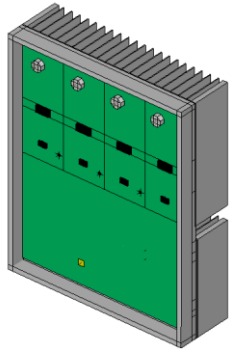


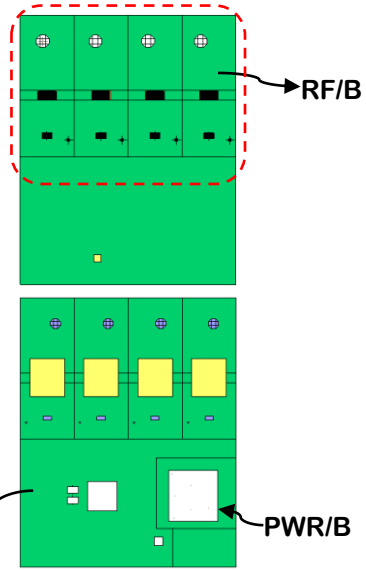
Boundary Conditions / Dimension / Structure

Exploded view:

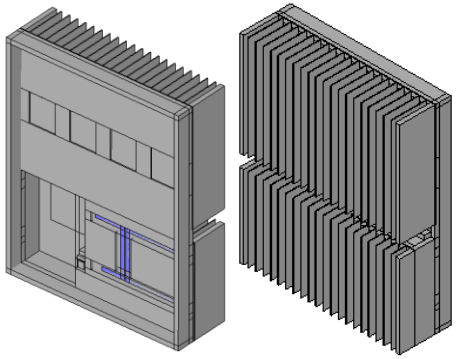


Cavity Filter:
- 339mm*272mm*42mm

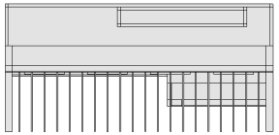


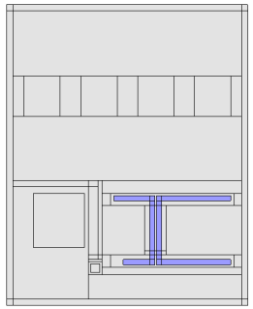


Digital/B + RF/B:320mm*256mm
PWR/B:130mm*95mm



Rear HSK





Rear HSK

Heat pipe:
2pcs D6 powder HP---FPGA

Base:
- thickness:7mm
- Al die-casting, Alloy (K=150W/mK)

Fin:
Bottom case-
- thickness:1.2mm
- Height:61mm(H), 22fin
- Fin pitch:12.8mm

Power Dissipation Overview

38.5W

Digital Board

FPGA (35W), Si5518 (2W), DDR (1.09W)

29W

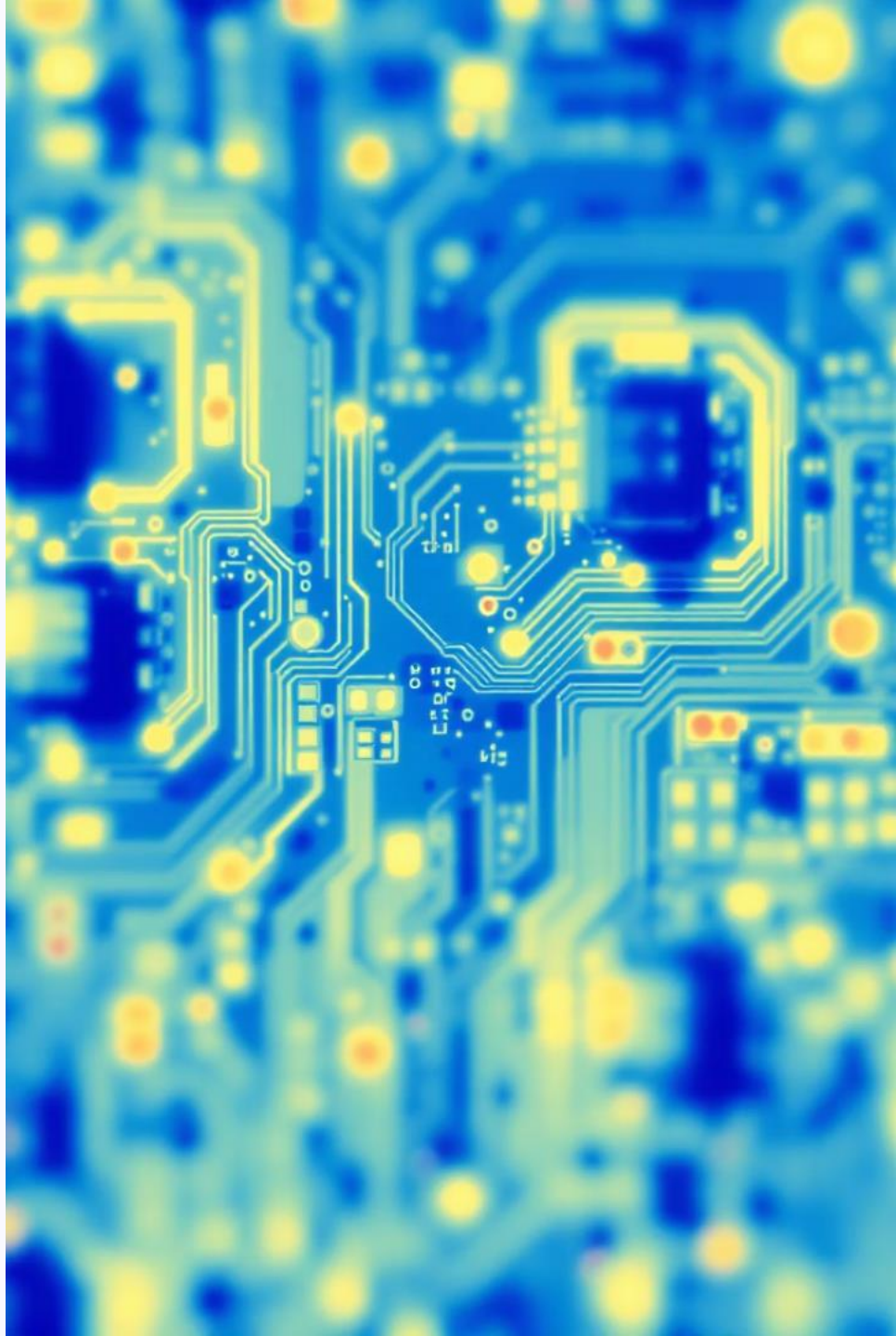
Power Module

286.12W

RF Board

Final amplifiers(211.96W), drivers(39W), Pre-drivers(1.472W), circulators(8.844W),Cavity filter(24.85W)

Total Pd:353.3W (@45℃ Backoff 1dB TDD 75%)



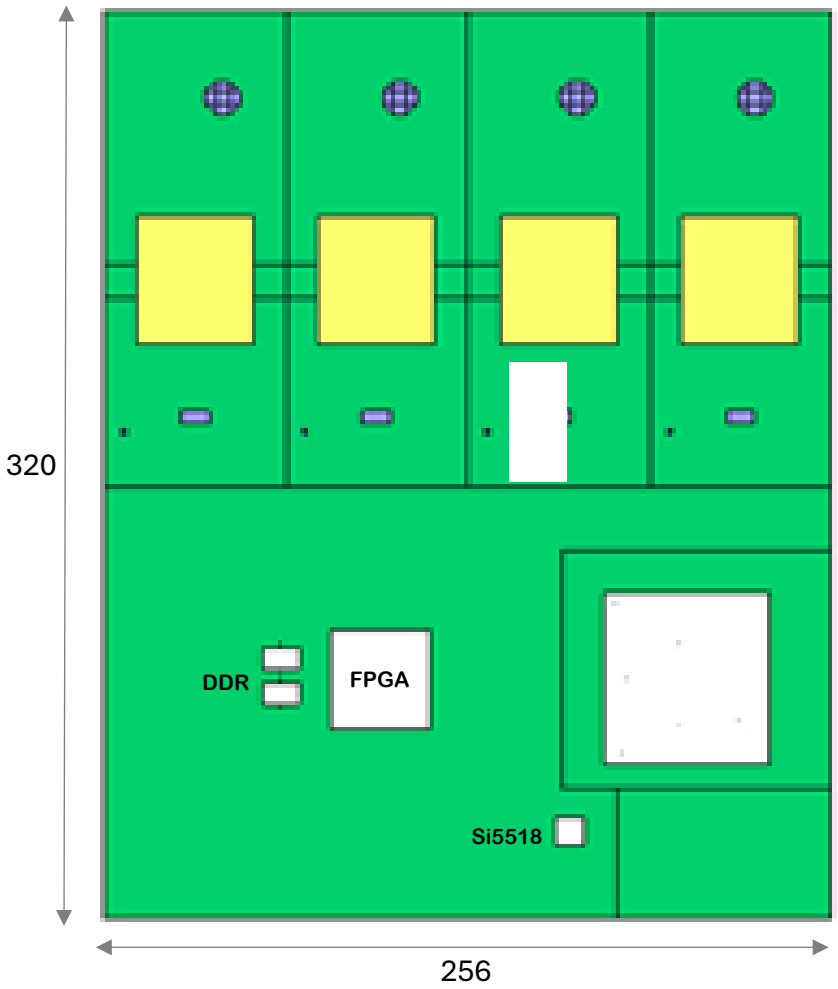
Power dissipation/ IC location /PCB placement

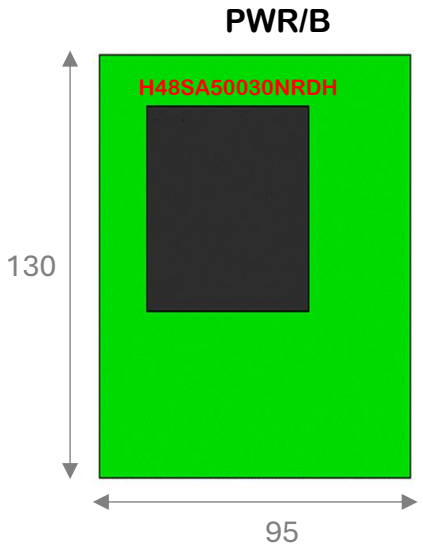
Digital/B:~38.09W(Pd)

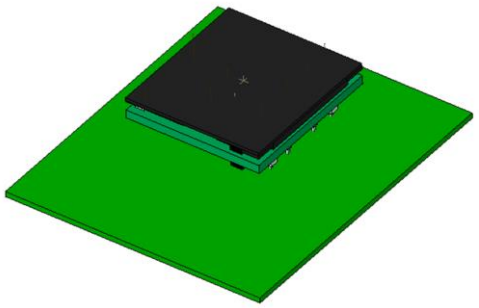
IC	Pd	Q'ty
FPGA (ffve1156_xczu67dr)	35W	1
Si5518	2W	1
DDR4	0.545W	1

Power module:~29W(Pd)

IC	Pd	Q'ty
H48SA50030NRDH	29W	1







Delta Confidential

Power dissipation/ IC location /PCB placement

RF:~286.12W(Pd)

IC	Pd	Q'ty
GTRB384608FC-Final	52.99W	4
B11G3338N81D-Driver	9.751W	4
BTS6201U-PreDriver	0.368W	4
Circulator	2.211W	4
Cavity filter	24.856W	1

Circulator property setting:

-- Material: Ferrite (鐵氧體)
--thermal conductivity:10W/mk
--Via:0.8mm/1.2mm, on 12mm*12mm area, K=17.7W/mk

B11G3338N81D-Driver property setting:

--Via: 8mil/16mm, on 9.8mm*4.8mm area, K=25.4W/mk

Symbol	Conditions		Value	Unit
$R_{th(j-c)}$	T_{case} = 90 °C			
	P_L = 5 W	[1]	1.7	K/W
	P_L = 10 W	[1]	1.4	K/W

T_j	junction temperature	[1]	-	200	°C

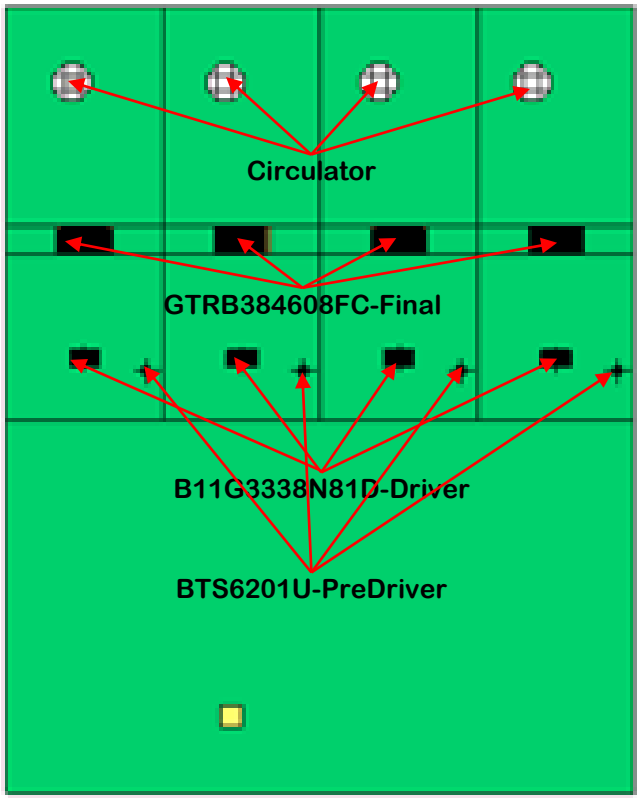
BTS6201U-Pre Driver property setting:

--Via: 8mil/16mm, on 1.6mm*1.6mm area, K=24.7W/mk

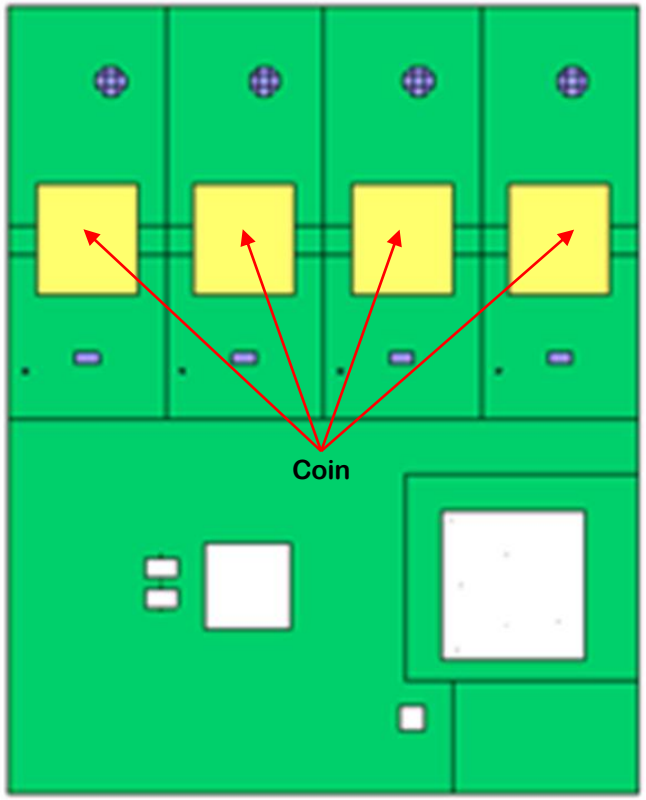
Table 8. Thermal characteristics

Symbol	Parameter	Conditions	Typ	Unit
$R_{th(j-case)}$	junction to case thermal resistance	[1] [2]	50	K/W

T_{case}	case temperature		-40	-	115	°C



PCB Top



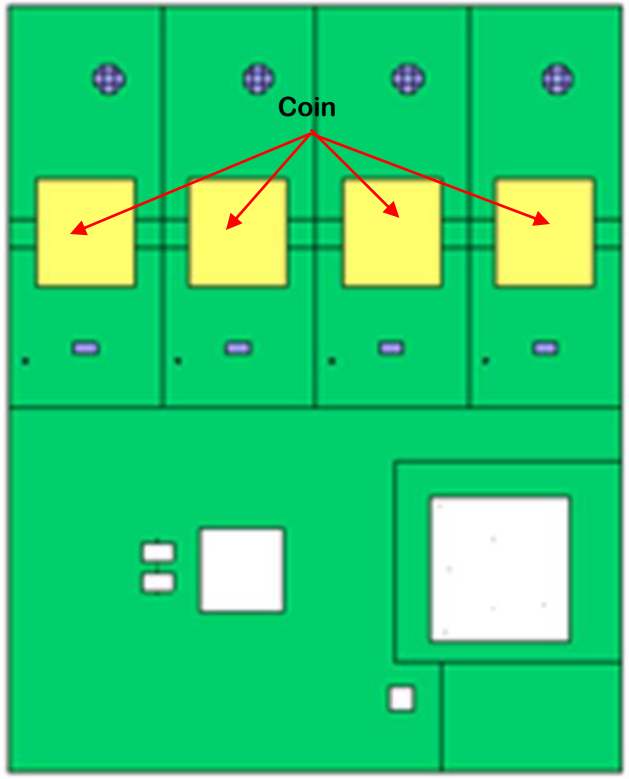
PCB Bottom

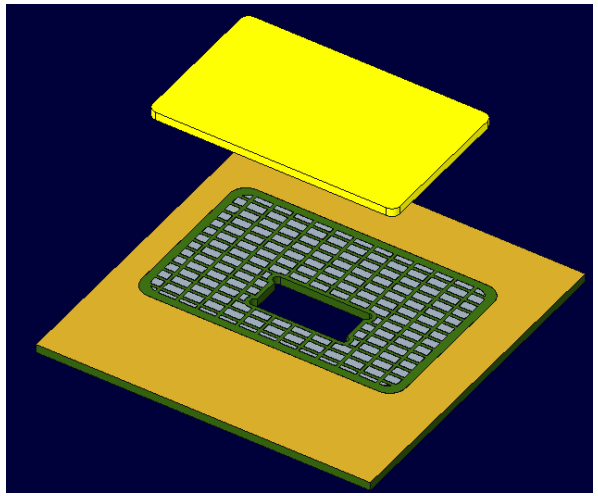
Power dissipation/ IC location /PCB placement

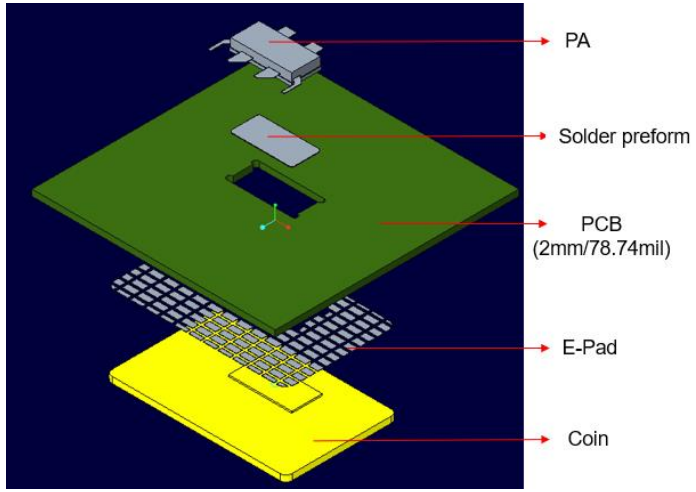
GTRB384608FC-Final property setting:

Thermal resistance (main)	T_{CASE} = 85°C, P_{DISS} = 91 W DC	°C/W	1.5

Product's qualification were performed at 225°C. Operation at T_J (275°C) reduces median time to failure.







Coin size:
45.5mm*41mm*3.18mm (2.5+0.68)
Weight:41.79g

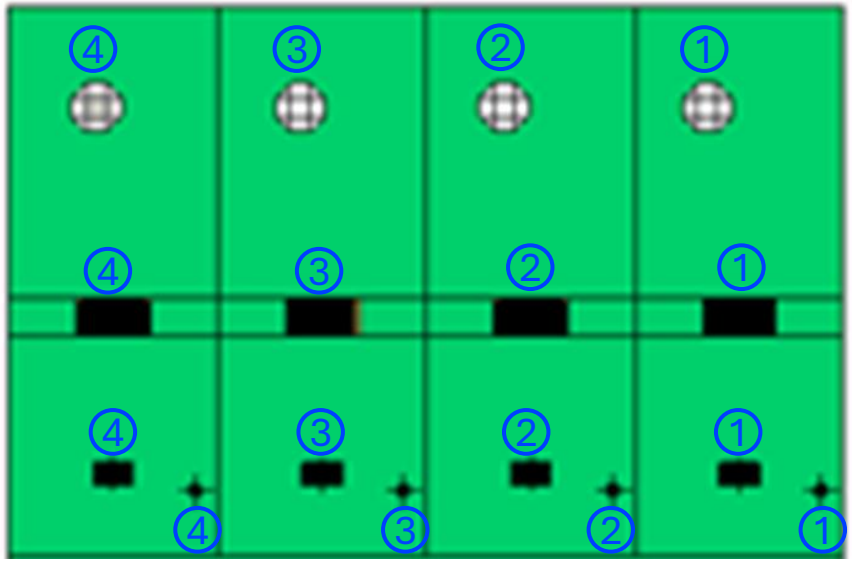
Simulation result (RF/B)

Ambient Temperature: 45 ˚C

External Velocity: 0m/s

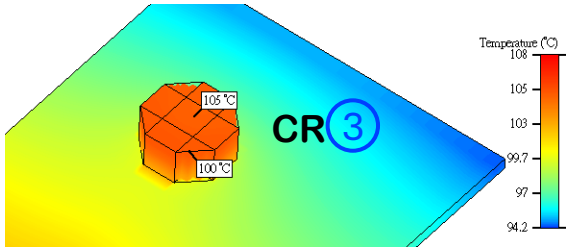
RF/B:~286.12W(Pd)

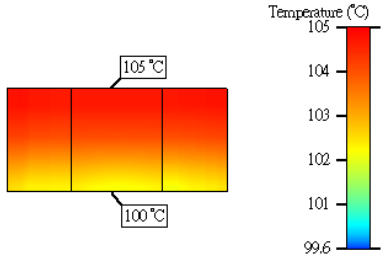
IC	Pd/pcs	Spec Tj/(Tc)	Simulation Tj/(Tc)	Margin
GTRB384608FC-Final-1	52.99W	225℃	187℃	17%
GTRB384608FC-Final-2	52.99W	225℃	190℃	16%
GTRB384608FC-Final-3	52.99W	225℃	190℃	16%
GTRB384608FC-Final-4	52.99W	225℃	187℃	17%
B11G3338N81D-Driver-1~4	9.751W	200℃	131℃~134℃	33%~34.5%
BTS6201U-PreDriver-1~4	0.368W	115℃	102℃~105℃	9%~12%
Circulator-1	4.6W	125℃	97~103℃	-
Circulator-2	4.6W	125℃	99.3~105℃	-
Circulator-3	4.6W	125℃	99.5~105℃	-
Circulator-4	4.6W	125℃	97.8~103℃	-



Circulator Temperature Characteristics& distribution

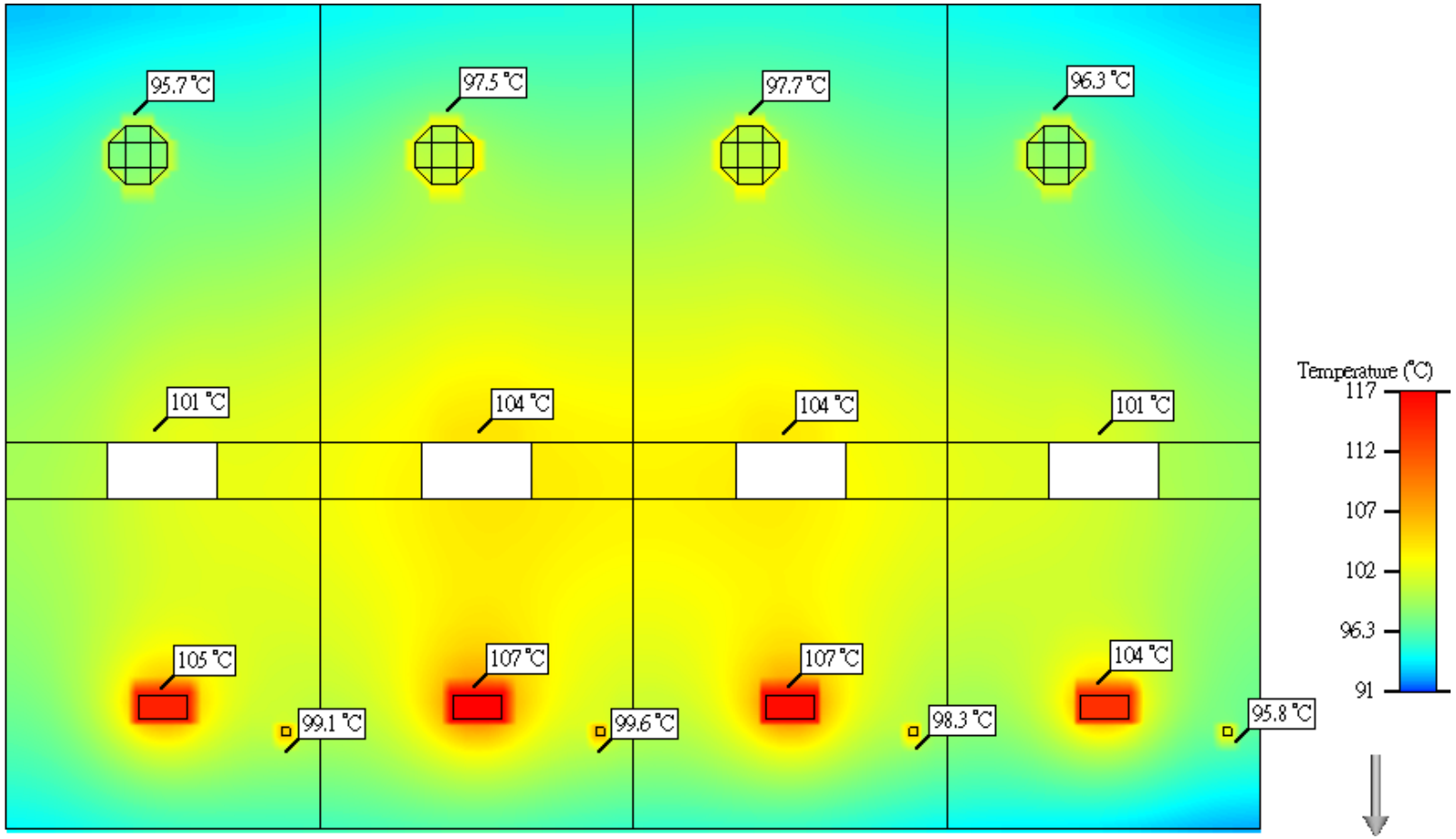
Return Loss(P1,P2,P3)	: 23 dB min(-40°C~+90°C)
	: 21 dB min(-40°C~+125°C)
Isolation(P1-P2,P2-P3)	: 23 dB min(-40°C~+90°C)
	: 21 dB min(-40°C~+125°C)
Insertion Loss (P1-P2,P2-P3)	: 0.30 dB Max(-40°C~+90°C)
	: 0.35 dB Max(-40°C~+125°C)





Simulation result (RF/B)

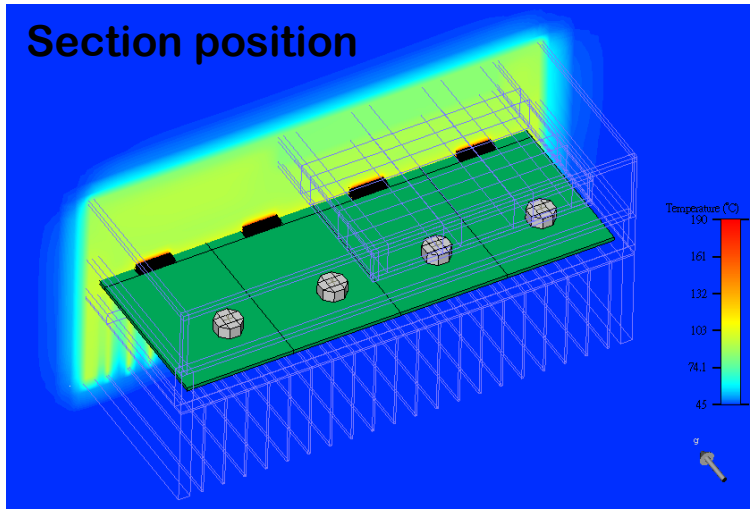
RF PCB TEMP

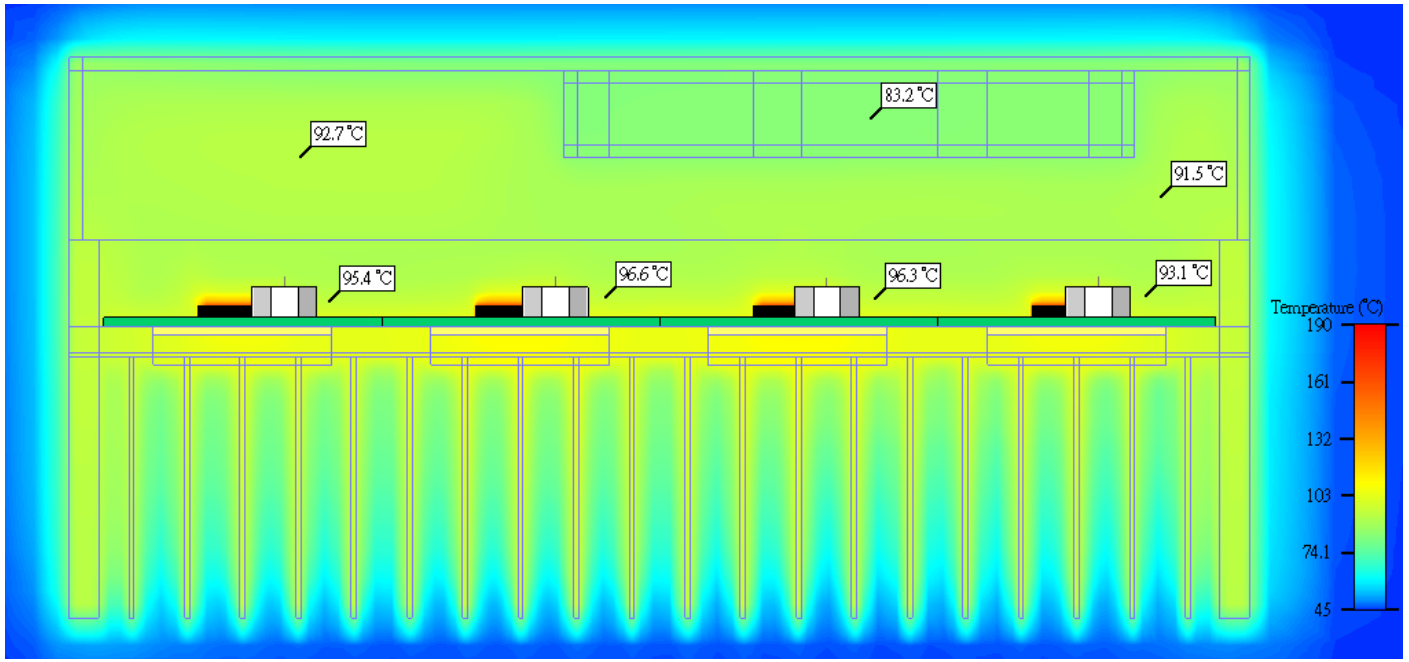


The RF PCB temperature is below 125°C, so there is low thermal risk for other passive components.

Simulation result (RF/B)

Air temperature near RF/B





Simulation result (Digital/B)

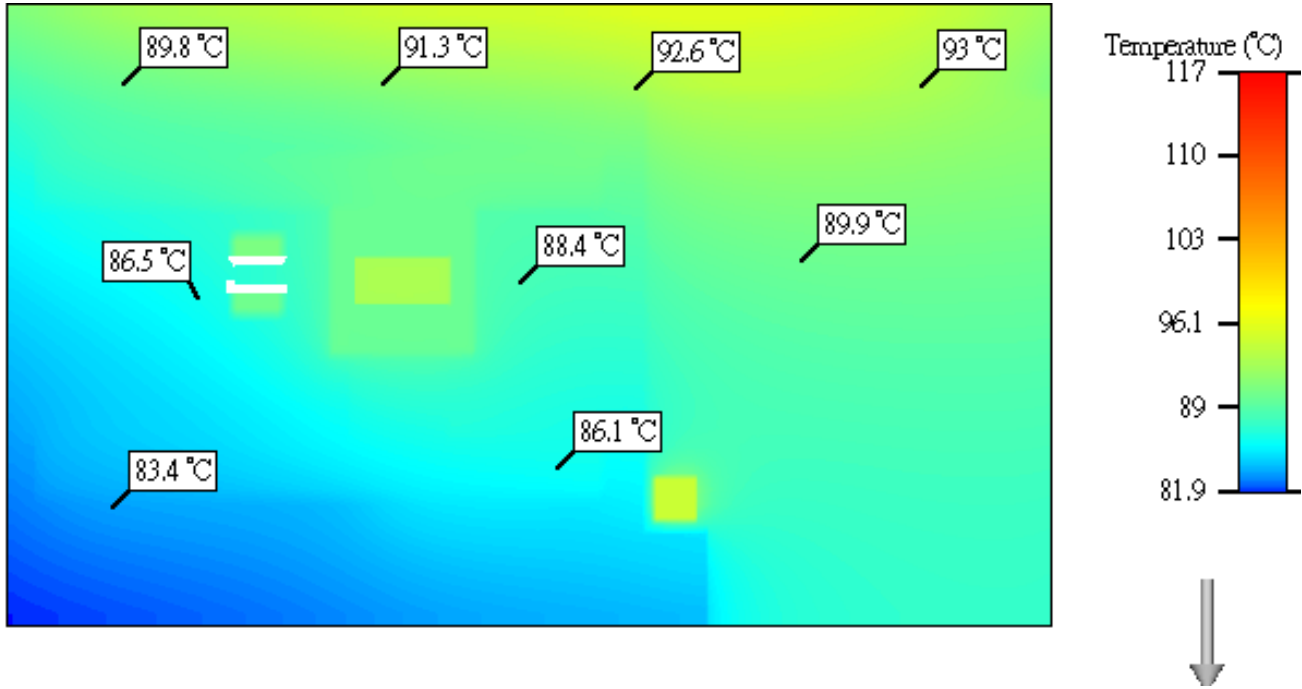
Ambient Temperature: 45 ˚C

External Velocity: 0m/s

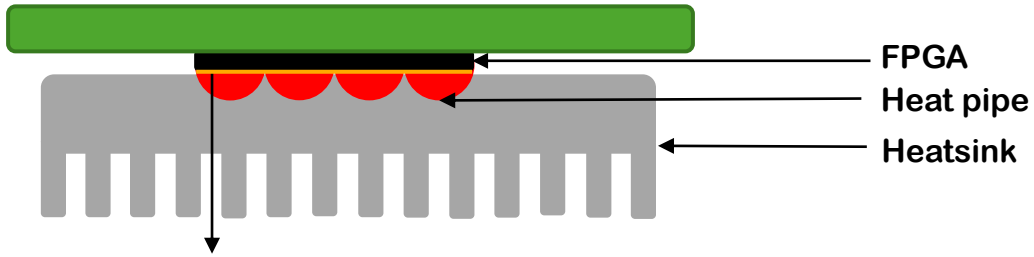
Digital/B:~38.09W(Pd)

IC	Pd	Spec Tj/(Tc)	Simulation Tj/(Tc)	Margin
FPGA	35W	100℃	97.9℃	2.1%
Si5518	2W	125℃	98.1℃	21.6%
DDR4	0.545W	95℃(Tc)	87.5℃	8%

PCB TEMP



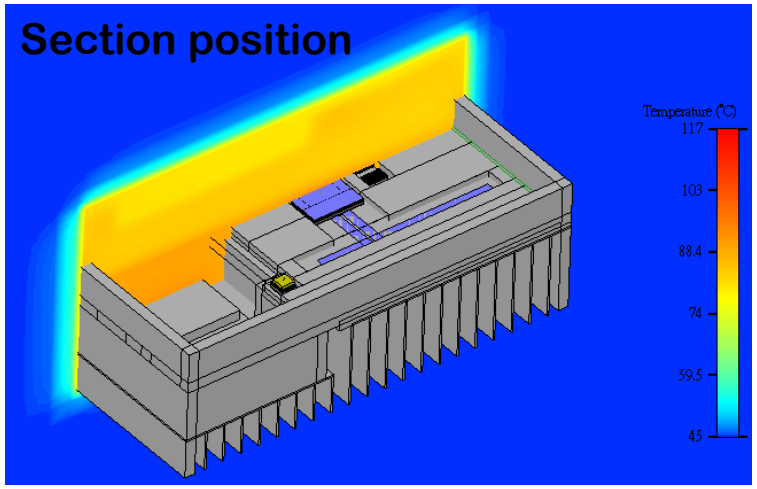
FPGA cooling stack

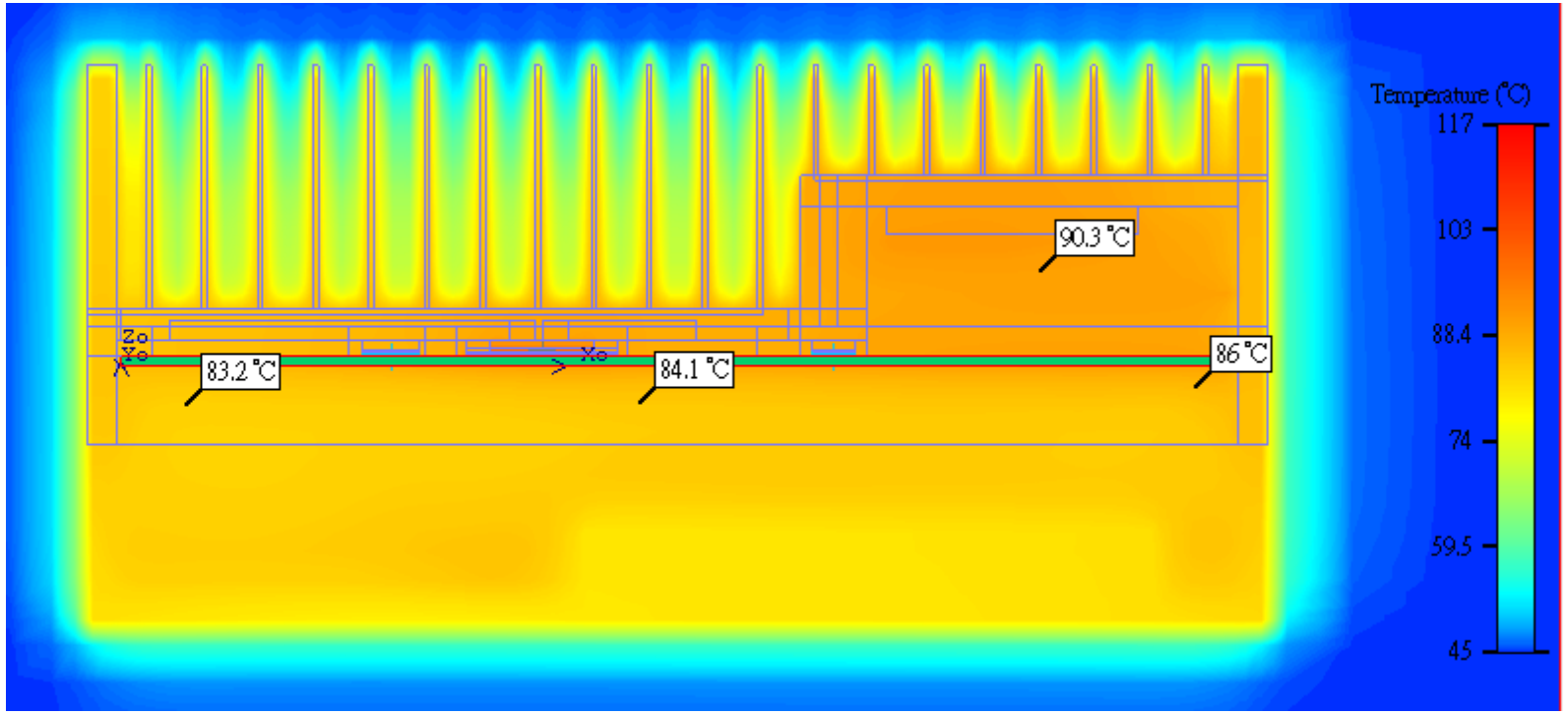


TIM:
Laird thermal putty 910 ,K=9.1W/mk
Thickness:0.4mm
The simulation evaluation a temperature difference of approximately 2˚C across both ends of the TIM.

Simulation result (Digital/B)

Air temperature near Digital/B





Simulation result (PWR/B)

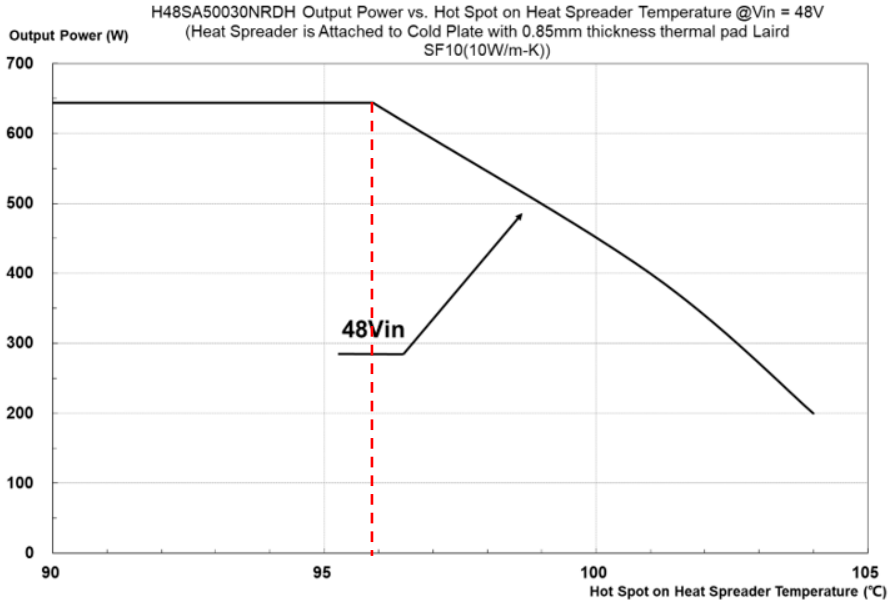
Ambient Temperature: 45 ˚C

External Velocity: 0m/s

Power module:~29W

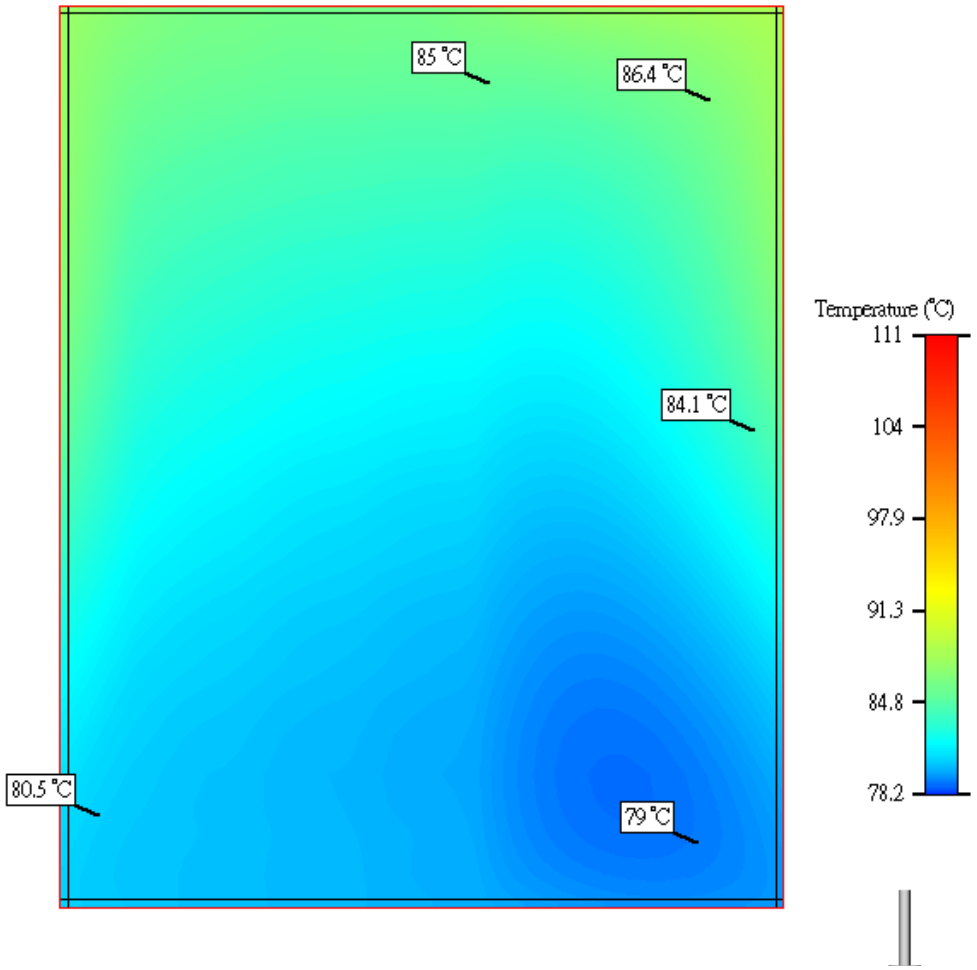
HSK	Pd	Spec Tc	Simulation Tj/(Tc)	Margin
H48SA50030NRDH	29W	96℃	91.5	4.7%

H48SA50030NRDH thermal curve

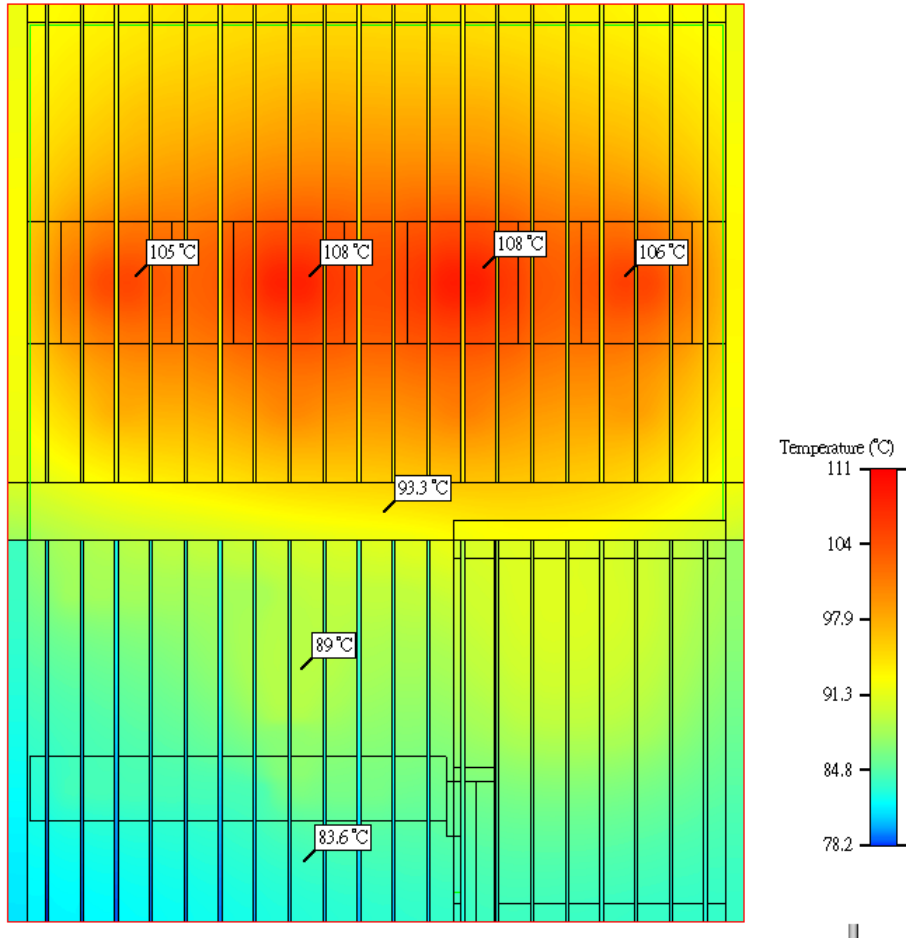


Temperature distribution

Ambient Temperature: 45 ˚C



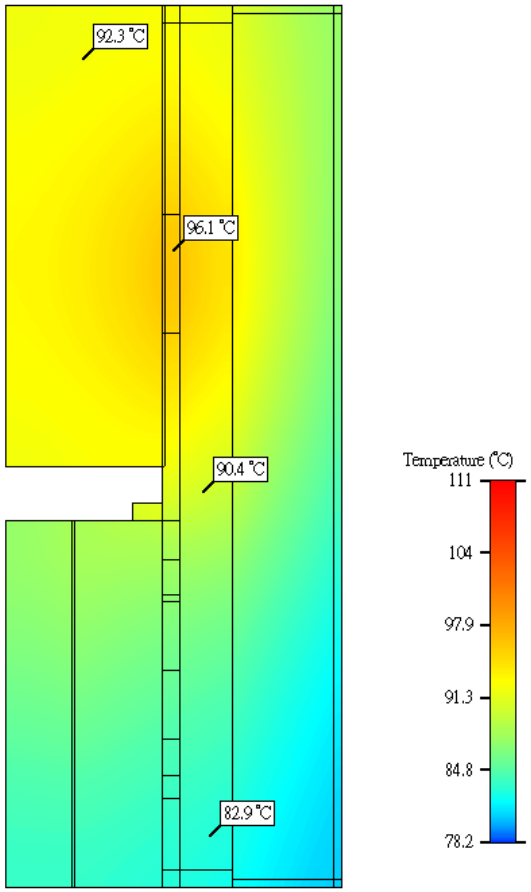
front view



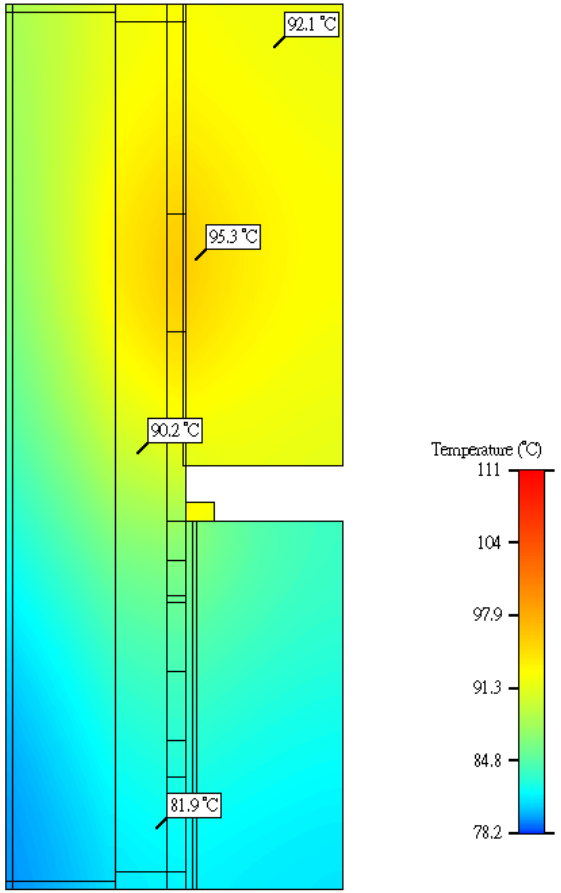
rear view

Temperature distribution

Ambient Temperature: 45 ˚C



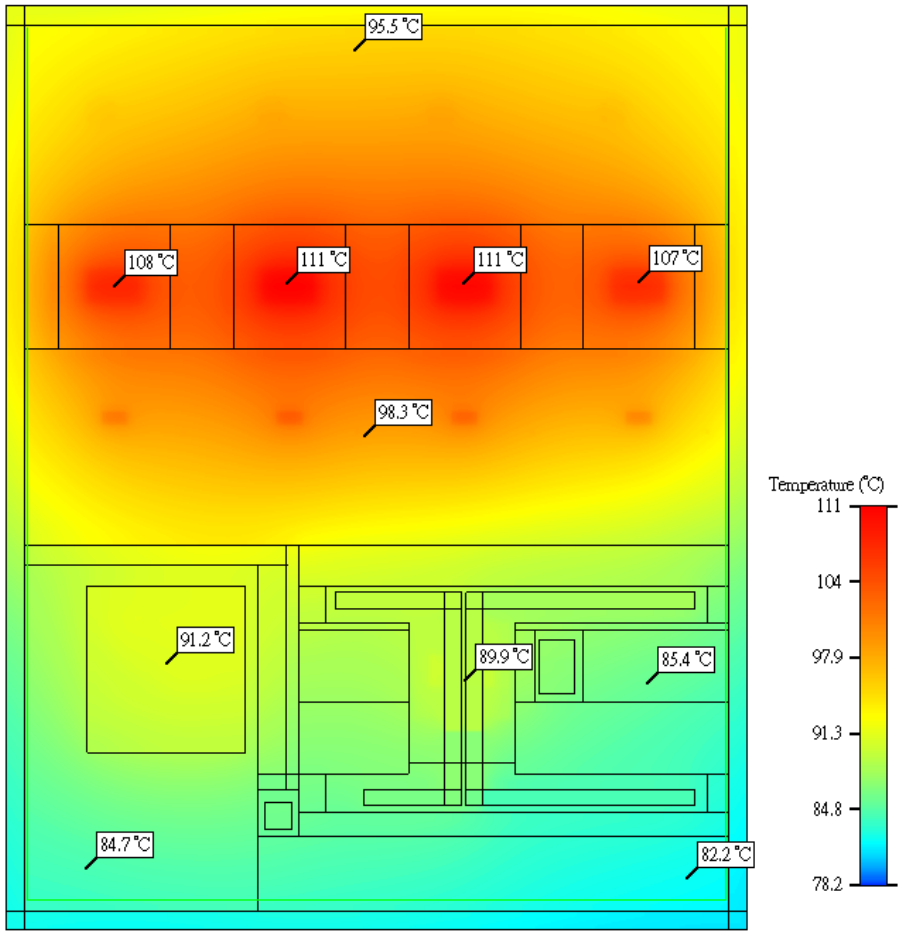
side view



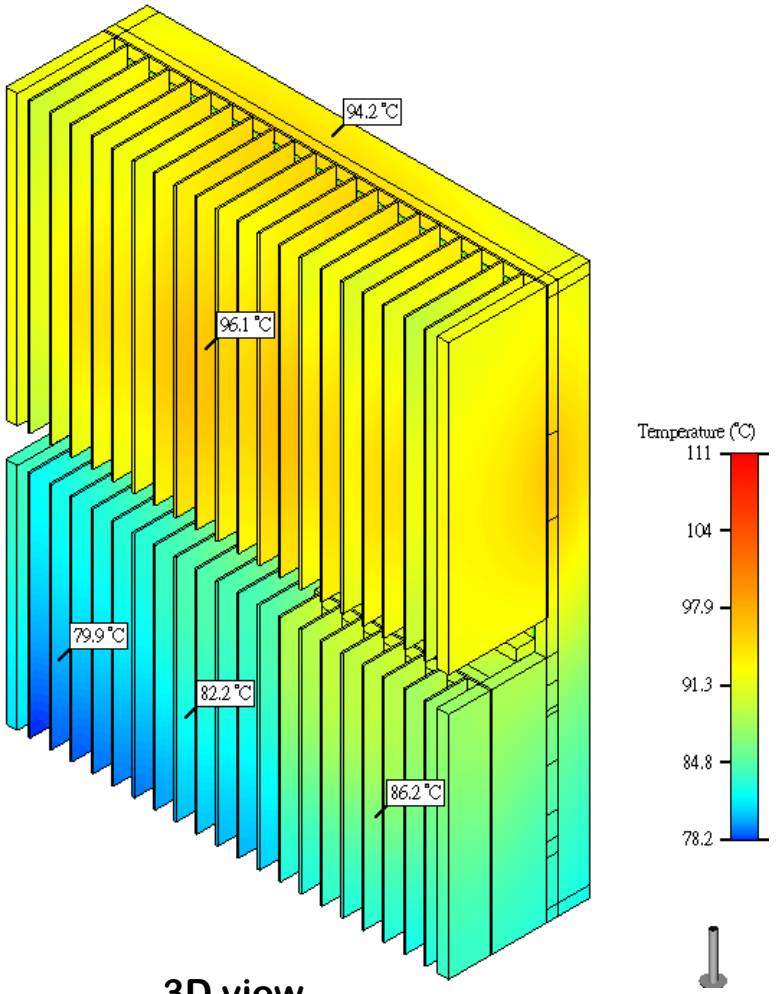
side view

Temperature distribution

Ambient Temperature: 45 ˚C



Rear HSK inside view



3D view

Smarter. Greener. Together.